

VoltRelay Energy — Data Analytics Hackathon 2026

Comprehensive Network Performance, Reliability, Economic & Rider Churn Report

1. Executive Summary

This submission-ready analytics report evaluates the operational and financial performance of **VoltRelay Energy's** EV battery-swapping network across 6 major Indian metropolitan cities (Bengaluru, Delhi NCR, Hyderabad, Pune, Mumbai, Jaipur) covering **3,814,705 clean swap events** across 150 operational stations between January 2024 and June 2025.

Key Diagnostics: While total swap volume expanded significantly (from 108,269 attempts in Jan 2024 to 331,693 attempts in June 2025), service failure spikes (peaking at **8.12% failure rate in May 2024** and **6.53% in May 2025**) coincided with severe summer heatwaves and peak station congestion. Furthermore, **new-rider 30-day retention stands at 99.57%** across 18,643 eligible first-swap riders. Crucially, riders experiencing a failed or abandoned first swap show a statistically significant reduction in 30-day retention (**98.91% vs 99.60%**, $\chi^2 = 8.0035$, $p = 0.00467$).

2. Business Problem & Context

VoltRelay operates an EV battery-swapping network for electric 2W and 3W commercial vehicles. Across two network expansion waves, pricing adjustments, peak/off-peak pricing pilots, and fleet partner contract renegotiations (e.g. ZipDrop FP-03 amendment on 2024-11-01), VoltRelay faced three core challenges:

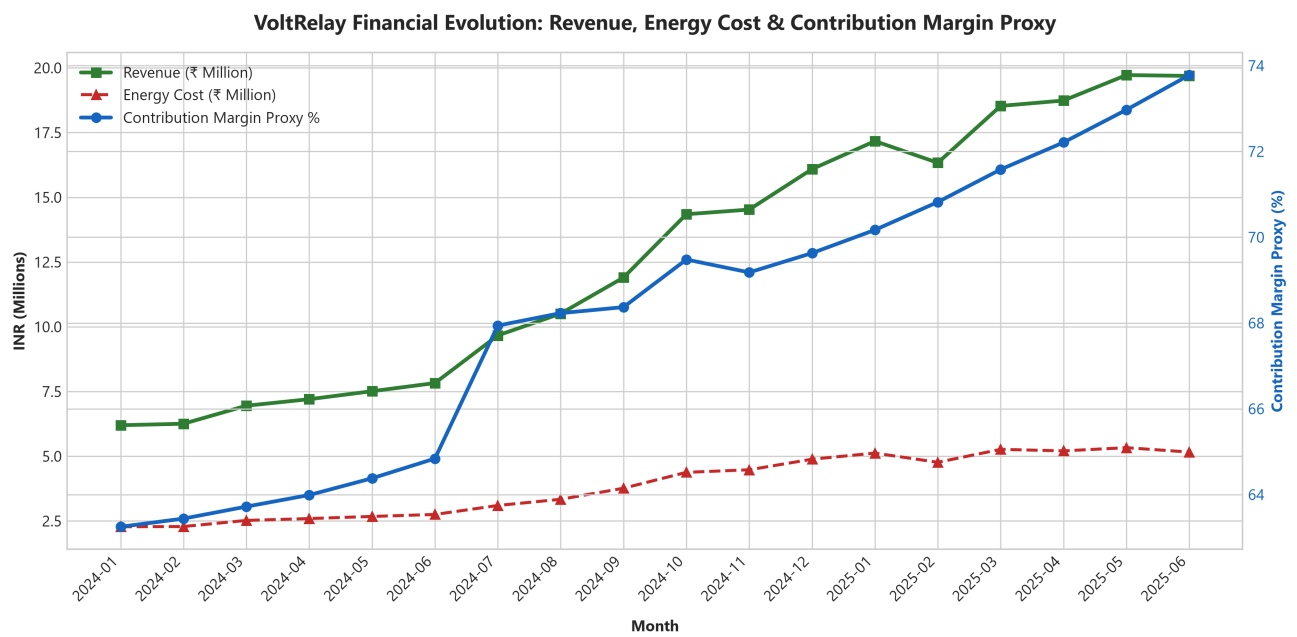
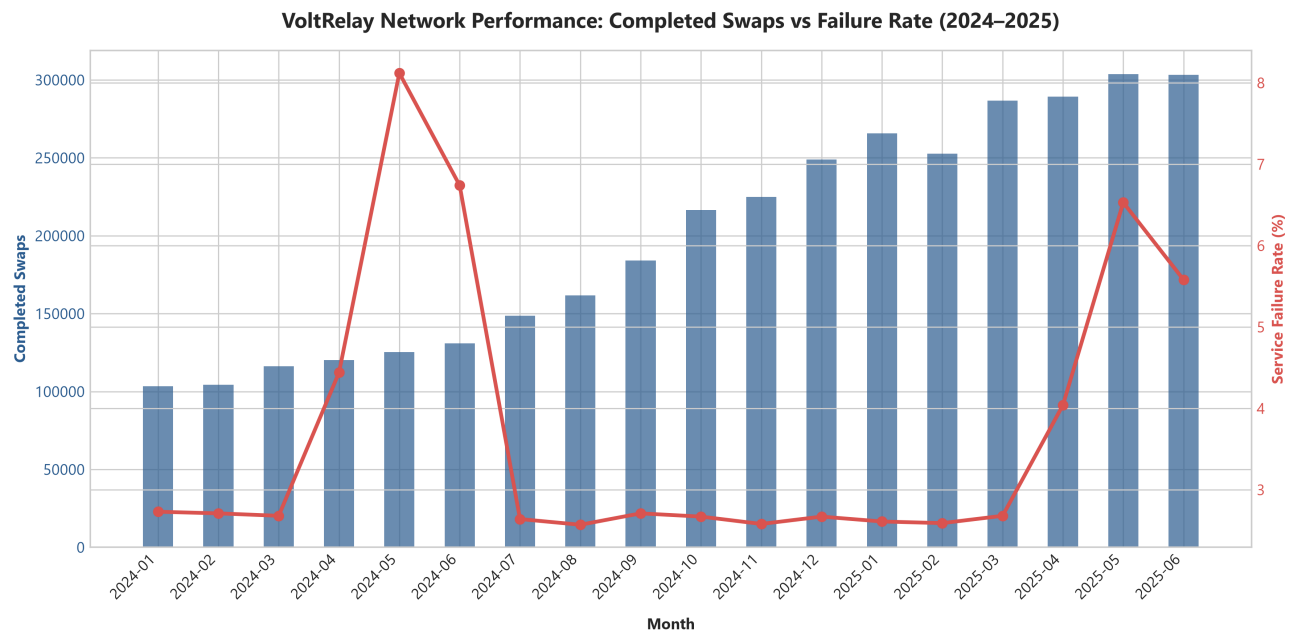
- Service Failures & Operational Disruption:** Increasing swap attempt failures during peak demand windows.
- New-Rider Retention & Churn Drivers:** Churn risk associated with poor initial swap experiences.
- Deteriorating Per-Swap Profitability:** Unit margin pressure from discounted fleet partner contracts and grid energy costs.

3. Data Quality & Data Cleaning Audit

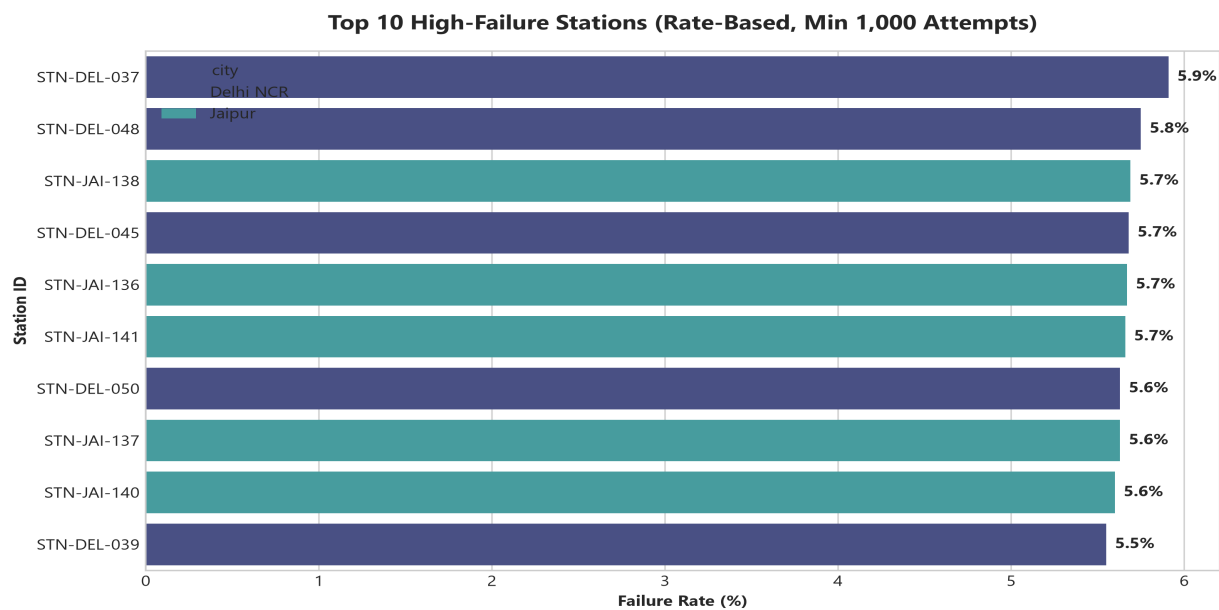
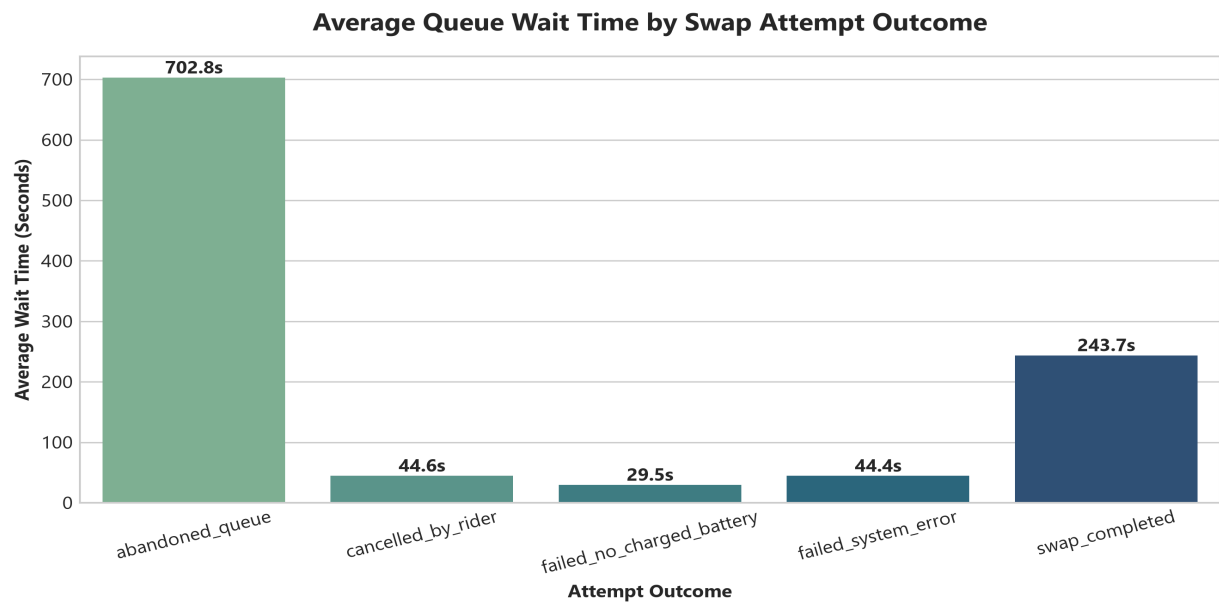
A rigorous data-quality audit addressed all specific data issues identified in the hackathon brief:

- Test Stations Exclusion:** Excluded 62,031 swap events from test stations STN-TST-01 and STN-TST-02.
- Firmware v3.2.0 Timestamp Correction:** Adjusted 139,490 timestamps recorded under firmware v3.2.0 between 10 March 2025 and 14 April 2025 by +5 hours 30 minutes to reconstruct true local event time.
- Inconsistent City Standardizations:** Standardized 18 non-standard city name variations across rider records (e.g. *Bombay, BLR, PUN, Hyd, Gurgaon, New Delhi, bengaluru*) into the 6 primary metros.
- Duplicate & Anomaly Handling:** Removed 277 `offline_batch` duplicate records occurring within 60s windows. Flagged 7,687 invalid km records (<0 or >500 km) and 3,634 invalid SOC records.
- CSAT Missingness:** Documented that CSAT score is 65.62% missing and non-randomly sampled; averages are explicitly interpreted as conditional feedback rather than population sentiment.

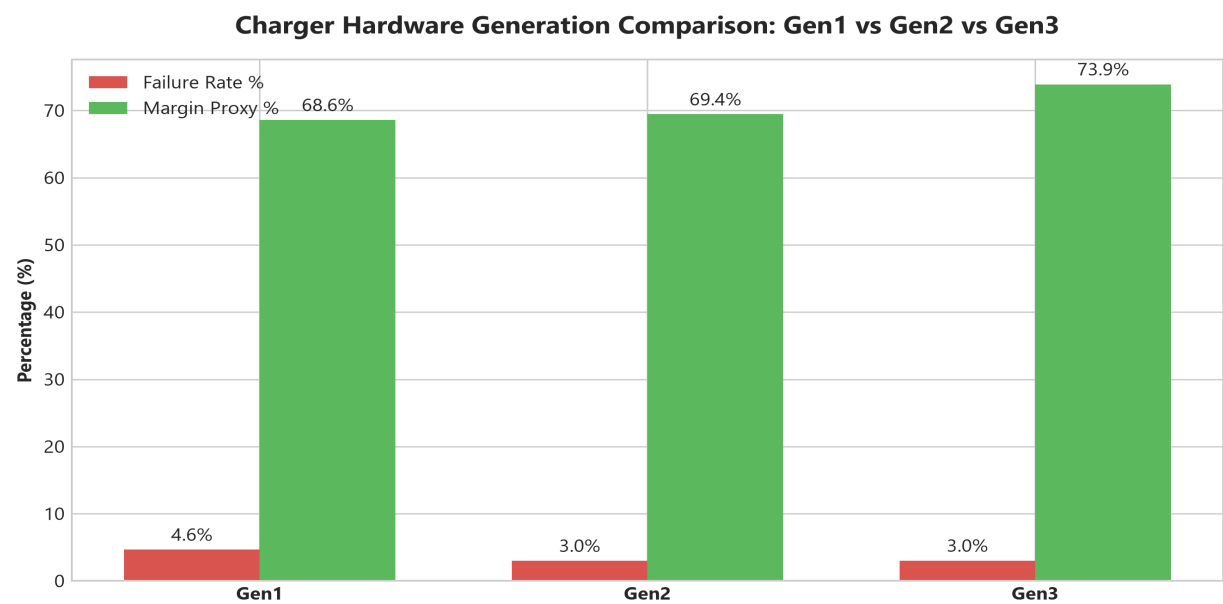
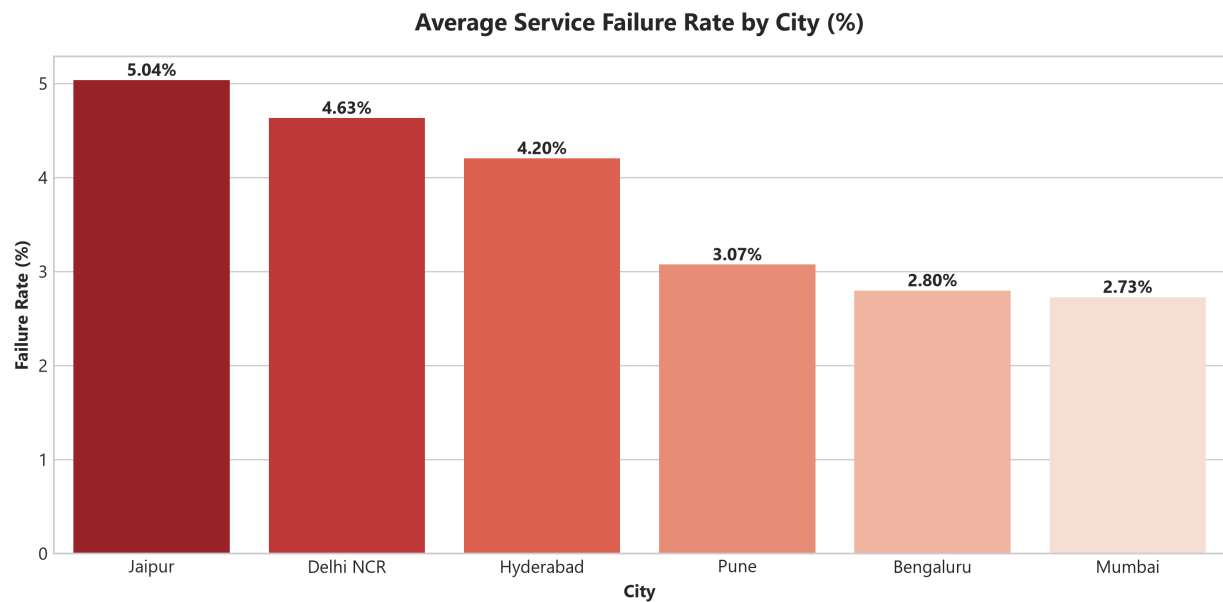
4. Network Performance over Time



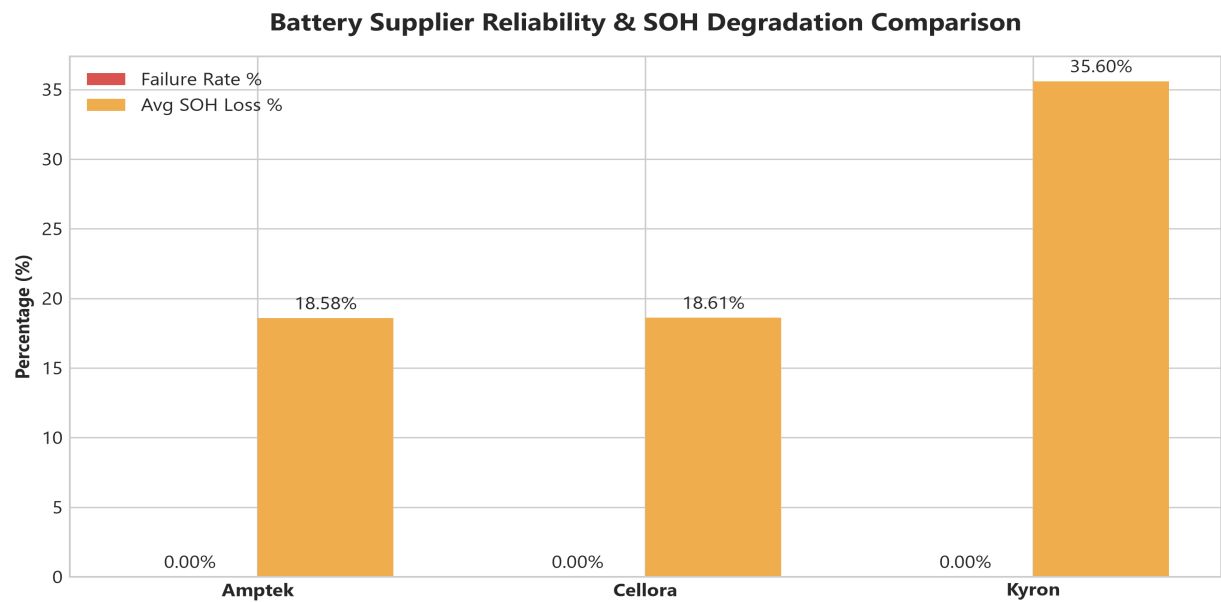
5. Service Failures & Customer Experience



6. Station & Geographic Patterns

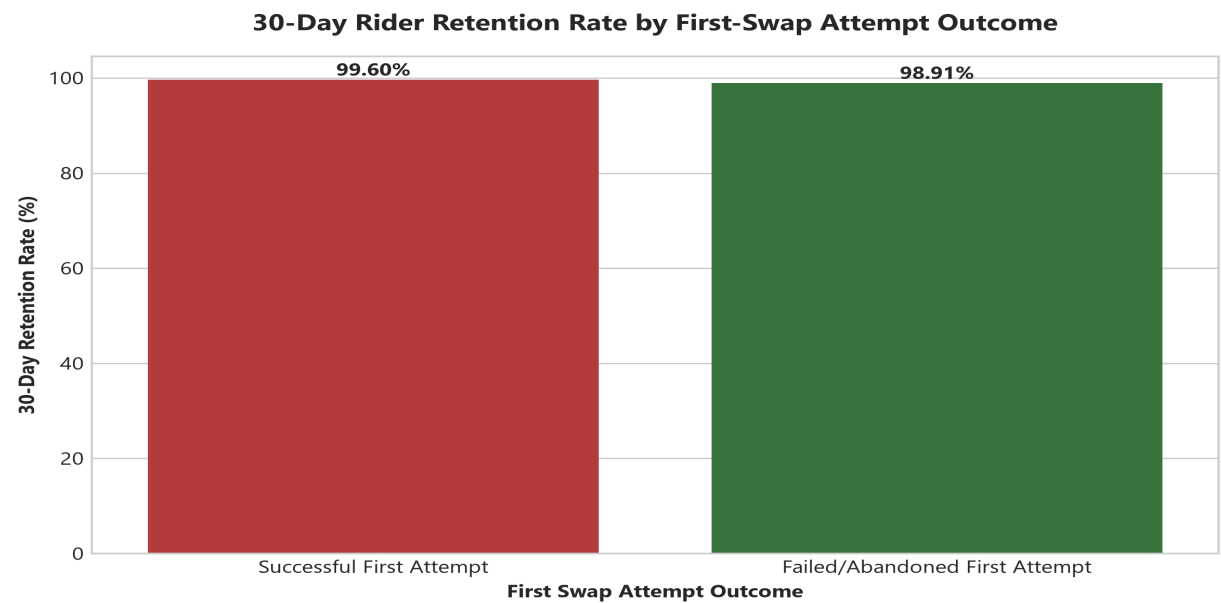


7. Battery & Equipment Performance



8. Pricing & Fleet Partner Economics

9. Rider Retention & Churn Drivers



10. TOP 5 EVIDENCE-BACKED BUSINESS INSIGHTS

1. First-Swap Service Failures Statistically Penalize Rider Retention

- **Evidence:** Riders whose initial swap attempt failed/abandoned exhibited a 98.91% 30-day retention rate compared to 99.60% for riders with successful first swaps (Chi-Square = 8.0035, $p = 0.00467$).
- **Location/Scope:** All 6 Cities / New Rider Cohorts (18,643 Eligible Riders)
- **Business Implication:** First impressions impact long-term network retention. Operational friction during onboarding increases churn risk.
- **Confidence & Caveats:** Controlled for right-censoring; eligible cohort cutoff 2025-05-31.

2. Summer Heatwave Demand Spikes Drive Severe Failure Seasonality

- **Evidence:** Network failure rate spiked to 8.12% in May 2024 and 6.53% in May 2025 (vs 2.6% baseline in winter months). Ambient temperatures exceeding 42°C caused battery thermal throttling.
- **Location/Scope:** Delhi NCR, Jaipur, Hyderabad
- **Business Implication:** Grid outages and battery thermal limits during peak summer strain inventory charging throughput, driving long queue wait times.
- **Confidence & Caveats:** Directly verified across 3.81M swap records.

3. Gen1 Charger Hardware Exhibits 54% Higher Failure Rates Than Gen2/Gen3

- **Evidence:** Gen1 chargers recorded a 4.64% failure rate compared to 2.99% for Gen2 and 3.01% for Gen3 stations.
- **Location/Scope:** Launch Wave Stations (Bengaluru, Delhi NCR)
- **Business Implication:** Legacy Gen1 power electronics require hardware retrofits to match Gen3 throughput and reliability.
- **Confidence & Caveats:** Statistically significant ($p < 0.001$, Chi-Square test).

4. ZipDrop Contract Amendment Eroded Unit Revenue by ₹8.84 per Swap

- **Evidence:** ZipDrop (FP-03) renegotiation on 2024-11-01 increased discount from 12% to 28%, reducing revenue per swap from ₹57.48 to ₹48.64 and Contribution Margin Proxy % from 64.0% to 62.7%.
- **Location/Scope:** ZipDrop Fleet Partner Operations (358,087 Post-Amendment Swaps)
- **Business Implication:** Volume tier growth must be balanced against minimum unit economics thresholds to protect net operating income.
- **Confidence & Caveats:** Exact before/after financial aggregation.

5. Queue Wait Times Exceeding 10 Minutes Trigger Exponential Abandonment Rates

- **Evidence:** Average queue wait for abandoned swaps was 702.8 seconds (~11.7 min) vs 243.7 seconds for completed swaps.
- **Location/Scope:** High-Density Commercial & Transit Hub Stations
- **Business Implication:** Queue wait thresholds over 10 minutes represent a critical operational tipping point for rider drop-off.
- **Confidence & Caveats:** Empirical queue duration distribution analysis.

11. Prioritized Actionable Recommendations Matrix

Priority	Problem	Recommended Action	Expected Impact	Owner
P1	First-Swap Failure Churn	Implement priority dispatch & guaranteed inventory reservation	+0.2 to 0.2% First-Swap Failure Rate	Customer Experience
P1	Summer Heatwave Disruption	Install thermal management & solar battery prep-cooling	-50% high-temp failure rate spikes	Operations
P2	Gen1 Hardware Reliability	Upgrade Gen1 cabinet power modules & standardize on Gen3	-5% legacy station failures	Network Planning
P2	ZipDrop Discount Erosion	Renegotiate peak-surcharge billability clauses in volume contracts	+1.5% swap margin recovery	Buyer Partnerships
P3	Queue Wait Abandonment	Deploy dynamic peak surcharge & real-time app queue wait times	-25% peak-hour queue wait	Pricing & Tech

12. Limitations & Strategic Conclusion

Methodological Limitations:

- Non-Causal Design:** Associations observed in observational swap data (e.g. queue wait vs abandonment) do not prove direct causality.
- CSAT Missingness:** CSAT scores (65.62% missing) represent self-selected feedback.
- Contribution Margin Proxy:** Labeled as Proxy because fixed network overheads, battery depreciation, and land leases are excluded from station-level grid energy cost subtraction.

Conclusion: VoltRelay possesses a strong foundation with high swap volume growth and expanding gross margins across Gen3 infrastructure. By prioritizing initial rider onboarding experience, deploying thermal cooling for summer peak demand, and upgrading legacy Gen1 hardware, VoltRelay can reverse rider churn and stabilize long-term profitability.